ABSTRACT OF THE DISCLOSURE

A system and method for representing fractional numbers in a digital computing system with high mathematical accuracy, avoiding the inherent imprecision of standard floating-point representations. A fractional number is represented as a composite N-gram data structure. The integer part of the number is represented using a parametric series representation, while the decimal part is represented as a sum of scaled members from a known, convergent mathematical series, such as the series from the Basel problem. The representation of the decimal part is made highly efficient by encoding each term as a pair of indices into two predefined matrices: a matrix of series members and a matrix of multipliers. This "geo-indexing" approach allows for the transmission of the "ingredients" for a high-precision decimal number in a highly compressed format.